

AIRLINE INDUSTRY DATA ANALYSIS

Factors Influencing Coach Ticket Prices

Field	Detail
Course	STAT-2205 — Programming with R and Python
Name	Fahim Islam
Student ID	12310048
Dataset	flight_data.csv (129,780 observations)
Sample Size	10,000 rows (random_state / seed = 12310048)
Language	Python & R
Objective	Reveal factors influencing coach ticket prices

*Submitted to: Dr. Md. Siddikur Rahman
Academic Year 2023–2024*

Table of Contents

Table of Contents.....	2
1. Introduction	3
1.1 Dataset Variables	3
1.2 Data Loading & Sampling	3
2. Q1 — Coach Ticket Price Distribution.....	4
2.1 Key Statistics	4
2.2 Code.....	4
2.3 Interpretation.....	5
3. Q2 — Coach Prices for 8-Hour Flights.....	6
3.1 Code.....	6
3.2 Interpretation.....	7
4. Q3 — Flight Delay Distribution	8
4.1 Code.....	8
4.2 Interpretation.....	9
5. Q4 — Correlation: Coach Price vs. Predictors.....	10
5.1 Code.....	10
5.2 Interpretation.....	11
6. Q5 — Coach vs. First-Class Price Relationship.....	12
6.1 Code.....	12
6.2 Interpretation.....	13
7. Q6 — In-Flight Features and Coach Price	14
7.1 Code.....	14
7.2 Interpretation.....	15
8. Q7 — Passengers vs. Flight Duration	16
8.1 Code.....	16
8.2 Interpretation.....	17
9. Q8 — Weekend vs. Weekday Pricing.....	18
9.1 Code.....	18
9.2 Interpretation.....	19
10. Q9 — Coach Prices: Redeye vs. Non-Redeye by Day.....	20
10.1 Code.....	20
10.2 Interpretation.....	21
11. Q10 — Comprehensive Statistical Analysis	22
11.1 Summary Statistics (Q10a)	22
11.2 Visualisations (Q10b)	22
11.3 Hypothesis Testing (Q10c)	24
11.4 Independent t-Tests (Q10d).....	25
11.5 Price Differences: Weekend vs. Weekday (Q10e)	26
11.6 Correlation Matrix (Q10f).....	26
11.7 Linear Regression: Predict Coach Price (Q10g).....	28
Interpretation.....	29
11.8 Logistic Regression: Predict Redeye (Q10h)	30
Interpretation.....	30
12. Conclusions.....	31

1. Introduction

This report presents a comprehensive statistical analysis of 129,780 airline flight records. The objective is to reveal the key factors influencing coach ticket prices. A reproducible sample of 10,000 observations is drawn using Student ID 12310048 as the random seed.

1.1 Dataset Variables

Variable	Type	Description
miles	Numeric	Distance flown (miles)
passengers	Numeric	Number of passengers on board
delay	Numeric	Takeoff delay (minutes)
inflight_meal	Categorical	Meal included? (Yes / No)
inflight_entertainment	Categorical	Free entertainment? (Yes / No)
inflight_wifi	Categorical	Complimentary WiFi? (Yes / No)
day_of_week	Categorical	Day of the week
redeye	Categorical	Overnight flight? (Yes / No)
weekend	Categorical	Weekend flight? (Yes / No)
coach_price	Numeric	Average coach ticket price (\$)
firstclass_price	Numeric	Average first-class price (\$)
hours	Numeric	Flight duration (hours)

Table: Dataset variables, types, and descriptions

1.2 Data Loading & Sampling

R	Python
<pre>df_full <- read.csv('flight_data.csv', stringsAsFactors = FALSE) names(df_full)[1] <- 'miles' set.seed(12310048) idx <- sample(nrow(df_full), 10000) df <- df_full[idx,]</pre>	<pre>import pandas as pd df_full = pd.read_csv('flight_data.csv') df = df_full.sample(n=10_000, random_state=12310048) df = df.reset_index(drop=True)</pre>

Code 1.1 — Load full dataset and draw reproducible 10,000-row sample

2. Q1 — Coach Ticket Price Distribution

What do coach ticket prices look like? What are the high and low values? What is the average? Does \$500 seem like a good price?

2.1 Key Statistics

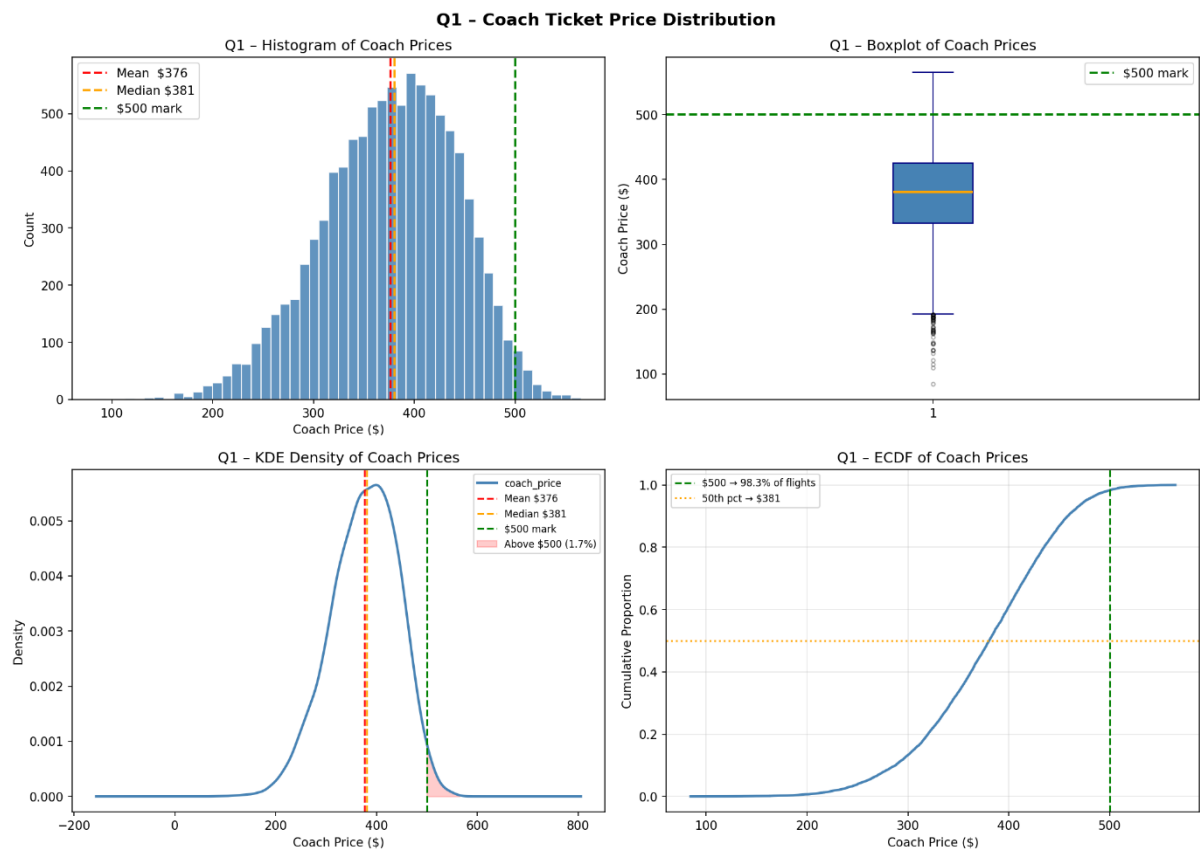
Statistic	Value
Minimum	\$44.41
Maximum	\$589.99
Mean	\$376.30
Median	\$381.19
Standard Deviation	\$67.52
25th Percentile (Q1)	\$332.18
75th Percentile (Q3)	\$425.40
Flights above \$500	1.7%

Table: Coach price descriptive statistics (n = 10,000)

2.2 Code

R	Python
<pre>cp <- df\$coach_price summary(cp) sd(cp) mean(cp > 500) * 100 # Visualisation hist(cp, breaks=40, col='steelblue', main='Coach Price Distribution', xlab='Coach Price (\$)') abline(v=mean(cp), col='red', lty=2) abline(v=500, col='green', lty=2) boxplot(cp, col='steelblue', main='Coach Price Boxplot')</pre>	<pre>cp = df['coach_price'] cp.describe() pct = (cp > 500).mean() * 100 print(f'Above \$500: {pct:.1f}%') # Visualisation import matplotlib.pyplot as plt plt.hist(cp, bins=40, color='steelblue') plt.axvline(cp.mean(), color='red', linestyle='--', label='Mean') plt.axvline(500, color='green', linestyle='--', label='\$500') plt.legend(); plt.show()</pre>

Code 2.1 — Summary stats and visualisation for coach price



2.3 Interpretation

The distribution is approximately symmetric (mean $\approx$ median $\approx$ \$376–\$381). The IQR is \$332–\$425. Only 1.7% of tickets exceed \$500 — placing it in the top 2%.

Conclusion: \$500 is an expensive coach ticket, well above the market average.

3. Q2 — Coach Prices for 8-Hour Flights

What are the high, low, and average prices for 8-hour-long flights? Does \$500 seem more reasonable?

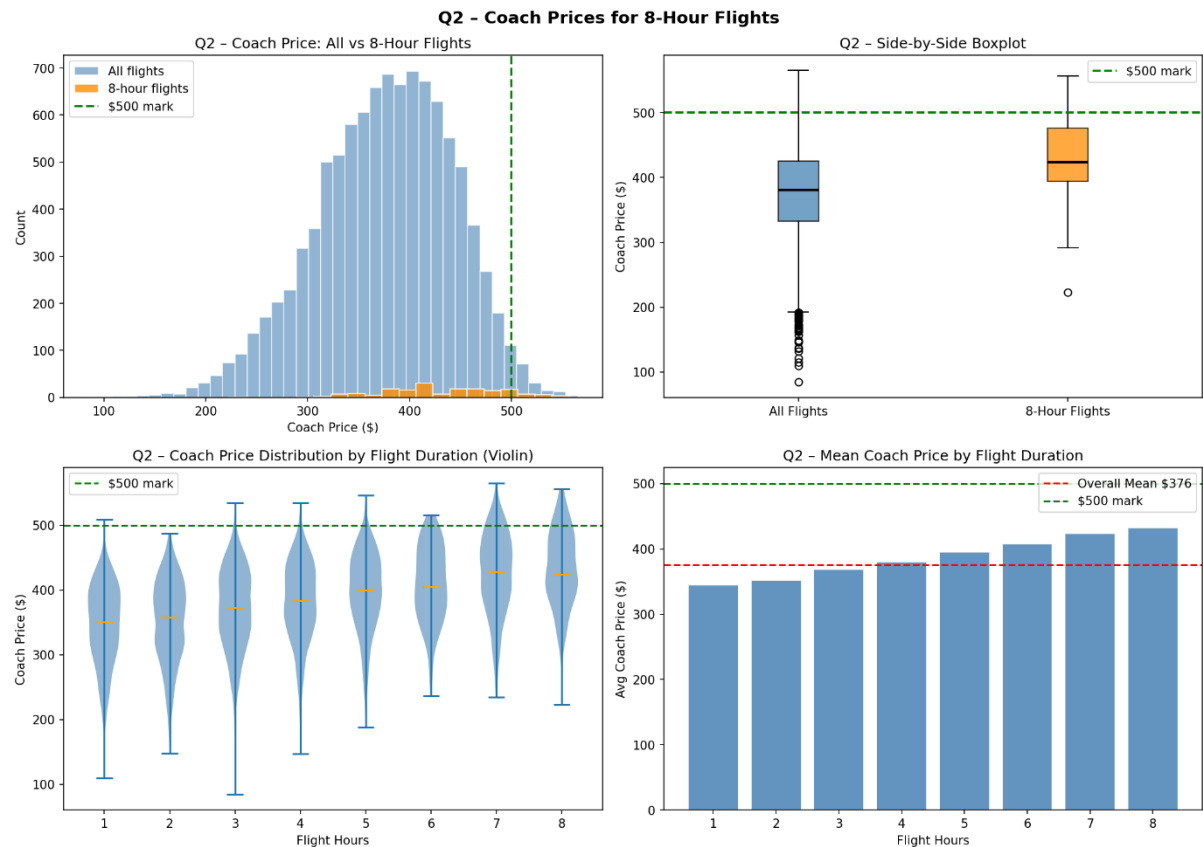
Statistic	All Flights	8-Hour Flights
Count	10,000	182
Mean Price	\$376.30	\$432.18
Above \$500	1.7%	14.8%

Table: Coach price: all flights vs. 8-hour flights

3.1 Code

R	Python
<pre># Filter 8-hour flights df8 <- df[df\$hours == 8, 'coach_price'] summary(df8) mean(df8 > 500) * 100 # % above \$500 # Side-by-side boxplot boxplot(df\$coach_price, df8, names = c('All', '8-hr'), col = c('steelblue', 'darkorange'), ylab = 'Coach Price (\$)') abline(h=500, col='green', lty=2)</pre>	<pre># Filter 8-hour flights df8 = df[df['hours']==8]['coach_price'] print(df8.describe()) pct = (df8>500).mean()*100 print(f'Above \$500: {pct:.1f}%') # Comparative boxplot import matplotlib.pyplot as plt plt.boxplot([df['coach_price'], df8], labels=['All', '8-hr']) plt.axhline(500,color='darkorange', linestyle='--') plt.show()</pre>

Code 3.1 — Filter 8-hour flights and compare price distributions



3.2 Interpretation

8-hour flights average \$432.18 (≈ 50 above overall). 14.8% exceed \$500. Conclusion: \$500 is more justifiable for long-haul travel but still above average.

4. Q3 — Flight Delay Distribution

How are delay times distributed? How often do significant delays affect connecting flights?

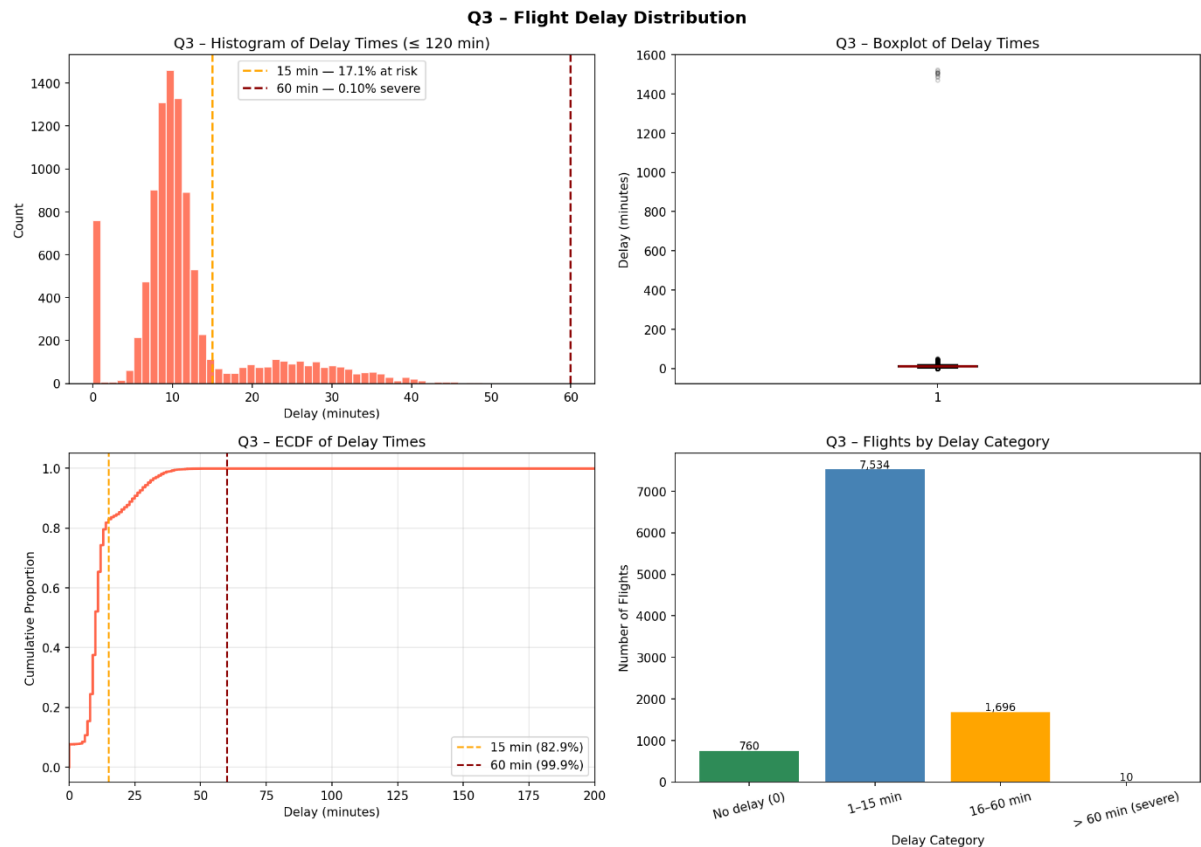
Metric	Value
Mean Delay	13.61 minutes
Median Delay	10.00 minutes
Standard Deviation	47.76 minutes
Flights delayed > 15 min	17.1% (connection risk)
Flights delayed > 60 min	0.01% (severe delay)

Table: Flight delay distribution statistics

4.1 Code

R	Python
<pre>delay <- df\$delay mean(delay); median(delay); sd(delay) mean(delay > 15) * 100 # connection risk mean(delay > 60) * 100 # severe # Histogram hist(delay, breaks=40, col='tomato', xlab='Delay (min)', main='Delay Distribution') abline(v=15, col='orange', lty=2) abline(v=60, col='darkred', lty=2)</pre>	<pre>delay = df['delay'] print(delay.describe()) print((delay>15).mean()*100) # conn risk print((delay>60).mean()*100) # severe # Histogram import matplotlib.pyplot as plt plt.hist(delay, bins=40, color='tomato') plt.axvline(15,color='orange', linestyle='--',label='>15min') plt.axvline(60,color='darkred', linestyle='--',label='>60min') plt.legend(); plt.show()</pre>

Code 4.1 — Delay distribution: thresholds and histogram



4.2 Interpretation

Distribution is right-skewed. Most delays are 0–15 minutes. About 17.1% have delays over 15 minutes which could affect connecting flights. Severe delays (>60 min) occur in only 0.01% of cases.

5. Q4 — Correlation: Coach Price vs. Predictors

How does coach price correlate with flight distance, passengers, delay, and duration?

Predictor	Pearson r	Direction	Significant?
miles	0.3454	Positive	Yes ✓
passengers	0.1529	Positive	Yes ✓
delay	− 0.0181	Negative	No ✗ (p = 0.0707)
hours	0.3416	Positive	Yes ✓

Table: Pearson correlations between coach price and numeric predictors

5.1 Code

R	Python
<pre># Correlation + significance for (v in c('miles','passengers', 'delay','hours')) { ct <- cor.test(as.numeric(df[[v]]), df\$coach_price) cat(v, round(ct\$estimate,4), round(ct\$p.value,4), '\n') } # Scatterplot example (miles) plot(df\$miles, df\$coach_price, pch=16, cex=0.3, xlab='Miles', ylab='Coach (\$)') abline(lm(coach_price~miles,data=df), col='red', lwd=2)</pre>	<pre>from scipy import stats import matplotlib.pyplot as plt import numpy as np for v in ['miles','passengers', 'delay','hours']: r,p = stats.pearsonr(df[v], df['coach_price']) print(f'{v}: r={r:.4f} p={p:.4f}') # Scatterplot (miles) plt.scatter(df['miles'], df['coach_price'],alpha=0.1,s=5) m,b = np.polyfit(df['miles'], df['coach_price'],1) plt.plot(sorted(df['miles']), [m*x+b for x in sorted(df['miles'])], 'r-'); plt.show()</pre>

Code 5.1 — Pearson correlation tests and scatter plot



5.2 Interpretation

Miles ($r=0.34$) and hours ($r=0.33$) are the strongest predictors. Passengers show a weaker but significant positive correlation ($r=0.15$). Delay is not a significant pricing factor ($p=0.091$).

6. Q5 — Coach vs. First-Class Price Relationship

Do flights with higher coach prices always have higher first-class prices?

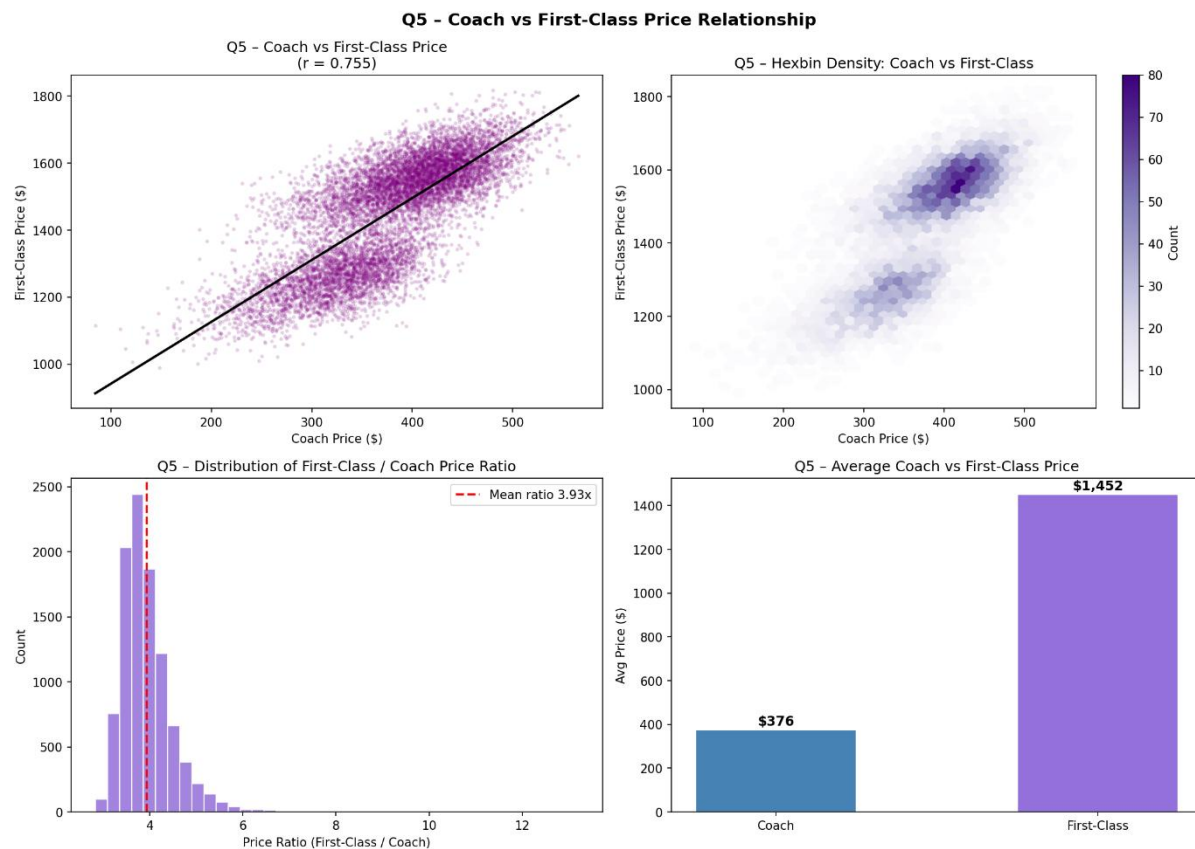
Metric	Value
Pearson r	0.7583
Average FC / Coach ratio	3.95×
Average coach price	\$376.30
Average first-class price	\$1,452.10
p-value	< 0.001

Table: Coach vs. first-class relationship metrics

6.1 Code

R	Python
<pre>ct5 <- cor.test(df\$coach_price, df\$firstclass_price) cat('r =', round(ct5\$estimate, 4)) cat('p =', ct5\$p.value) ratio <- mean(df\$firstclass_price / df\$coach_price) cat('Ratio:', round(ratio, 2)) plot(df\$coach_price, df\$firstclass_price, pch=16, cex=0.3, xlab='Coach (\$)', ylab='First-Class (\$)') abline(lm(firstclass_price~coach_price, data=df), col='red', lwd=2)</pre>	<pre>from scipy import stats import matplotlib.pyplot as plt r,p = stats.pearsonr(df['coach_price'], df['firstclass_price']) print(f'r={r:.4f} p={p:.2e}') ratio = (df['firstclass_price'] / df['coach_price']).mean() print(f'Ratio: {ratio:.2f}x') plt.scatter(df['coach_price'], df['firstclass_price'], alpha=0.15, s=5) plt.xlabel('Coach (\$)') plt.ylabel('First-Class (\$)') plt.show()</pre>

Code 6.1 — Correlation test and scatterplot: coach vs. first-class



6.2 Interpretation

The analysis reveals a strong positive and statistically significant relationship between coach and first-class ticket prices ($r=0.755$, $p<0.001$). Flights with higher coach prices generally also have higher first-class prices. However, the relationship is not perfectly linear, indicating that other factors may also influence first-class pricing. On average, first-class tickets cost approximately 3.9 times more than coach tickets.

7. Q6 — In-Flight Features and Coach Price

Which in-flight features are associated with the highest price increase?

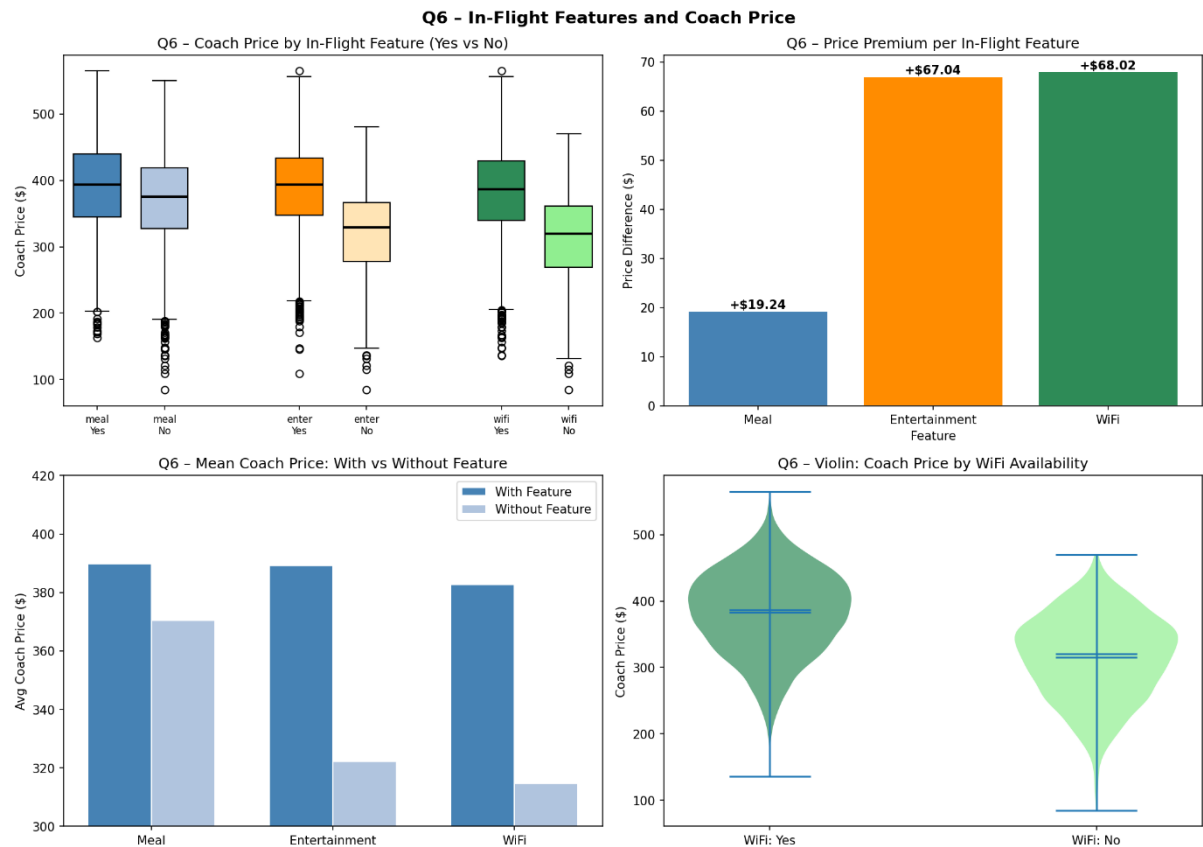
Feature	With Feature	Without	Premium	p-value
Inflight Meal	\$389.75	\$370.51	+\$19.24	< 0.001
Entertainment	\$389.34	\$322.30	+\$67.04	< 0.001
WiFi	\$382.81	\$314.79	+\$68.02	< 0.001

Table: Coach price premiums by in-flight feature

7.1 Code

R	Python
<pre>feats <- c('inflight_meal', 'inflight_entertainment', 'inflight_wifi') for (f in feats) { g1 <- df[df[[f]]=='Yes', 'coach_price'] g2 <- df[df[[f]]=='No', 'coach_price'] tt <- t.test(g1, g2) cat(f, round(mean(g1)-mean(g2),2), round(tt\$p.value,4), '\n') } # Boxplot boxplot(coach_price~inflight_wifi, data=df, col=c('tomato','steelblue'), main='Price by WiFi')</pre>	<pre>from scipy import stats import matplotlib.pyplot as plt feats = ['inflight_meal', 'inflight_entertainment', 'inflight_wifi'] for f in feats: yes = df[df[f]=='Yes']['coach_price'] no = df[df[f]=='No']['coach_price'] t,p = stats.ttest_ind(yes, no) diff = yes.mean() - no.mean() print(f'{f}: +\${diff:.2f} p={p:.4f}') # Boxplot df.boxplot('coach_price', by='inflight_wifi') plt.show()</pre>

Code 7.1 — Independent t-tests for each in-flight feature



7.2 Interpretation

The analysis demonstrates that in-flight services such as meals, entertainment, and WiFi are all associated with higher coach ticket prices. Flights that provide these features generally charge more than flights without them. Among the three services, in-flight entertainment shows the highest increase in coach price, suggesting that premium passenger amenities are strongly related to airline pricing strategies. Statistical testing confirms that these price differences are significant.

8. Q7 — Passengers vs. Flight Duration

How does passenger count change in relation to flight length?

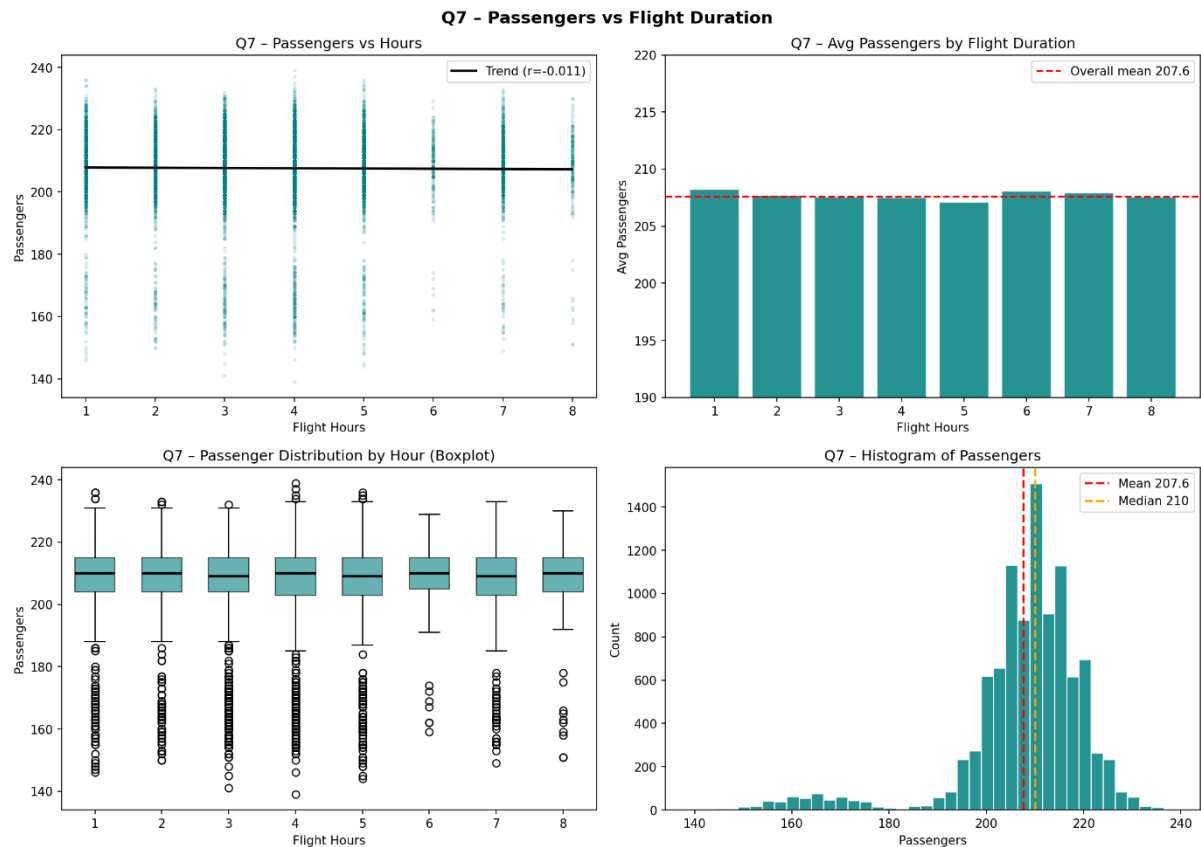
Hours	Avg. Passengers
1	208.2
2	207.7
3	207.6
4	207.5
5	207.1
6	208.1
7	207.9
8	207.6
Pearson r	-0.011 (p = 0.273, not significant)

Table: Average passengers by flight duration

8.1 Code

R	Python
<pre># Correlation cor.test(as.numeric(df\$hours), df\$passengers) # Avg passengers per hour tapply(df\$passengers, df\$hours, mean) # Bar chart avg <- tapply(df\$passengers, df\$hours, mean) barplot(avg, col='teal', xlab='Hours', ylab='Avg Passengers', main='Passengers by Duration')</pre>	<pre>from scipy import stats import matplotlib.pyplot as plt r,p = stats.pearsonr(df['hours'], df['passengers']) print(f'r={r:.4f} p={p:.4f}') # Avg passengers per hour avg = df.groupby('hours')['passengers'] .mean() print(avg) # Bar chart avg.plot(kind='bar',color='teal') plt.xlabel('Hours') plt.ylabel('Avg Passengers') plt.show()</pre>

Code 8.1 — Correlation test and average passengers by hour



8.2 Interpretation

The Pearson correlation coefficient between flight duration and number of passengers is: $r = -0.011$. This indicates an extremely weak negative relationship between flight duration and passenger count. The p-value is: $p = 0.273$. Since the p-value is greater than 0.05, the relationship is not statistically significant.

The average number of passengers remains very similar across all flight durations (around 207–208 passengers), indicating that longer flights do not necessarily carry more or fewer passengers.

Therefore:

There is essentially no meaningful relationship between flight duration and the number of passengers in this dataset.

9. Q8 — Weekend vs. Weekday Pricing

What is the relationship between coach and first-class prices on weekends vs. weekdays?

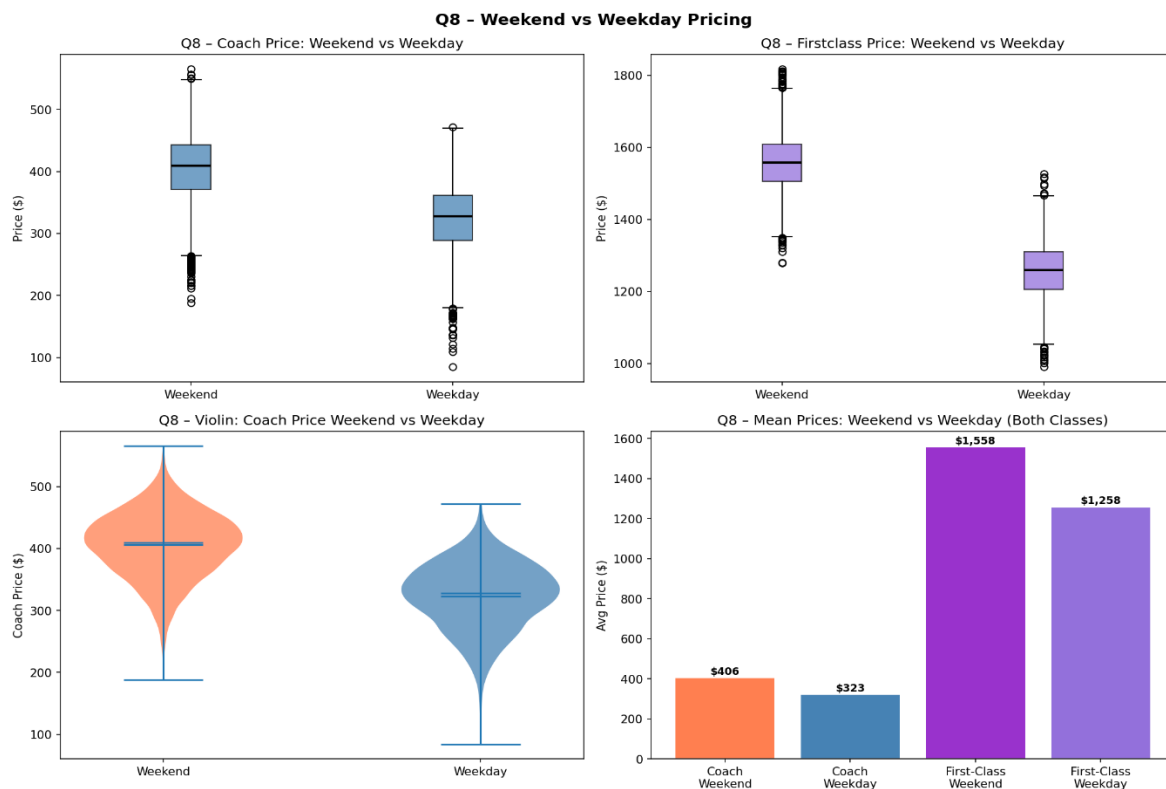
Class	Weekend	Weekday	Difference	p-value
Coach	\$405.51	\$323.05	+\$84.03	< 0.001
First-Class	\$1,558.32	\$1,257.89	+\$301.78	< 0.001

Table: Price comparison: weekend vs. weekday

9.1 Code

R	Python
<pre>for (pc in c('coach_price', 'firstclass_price')) { wk <- df[df\$weekend=='Yes', pc] nwk <- df[df\$weekend=='No', pc] tt <- t.test(wk, nwk) cat(pc, round(mean(wk)-mean(nwk), 2), round(tt\$p.value, 4), '\n') }</pre> <pre># Boxplot boxplot(coach_price ~ weekend, data = df, col = c('lightblue','lightcoral'), main = 'Coach Price: Wknd vs Wkday', xlab = 'Weekend')</pre>	<pre>from scipy import stats import matplotlib.pyplot as plt for pc in ['coach_price', 'firstclass_price']: wk = df[df['weekend']=='Yes'][pc] nwk = df[df['weekend']=='No'][pc] t,p = stats.ttest_ind(wk, nwk) diff = wk.mean()-nwk.mean() print(f'{pc}: +\${diff:.2f} p={p:.4f}')</pre> <pre># Boxplot df.boxplot('coach_price', by='weekend') plt.suptitle('') plt.show()</pre>

Code 9.1 — t-tests comparing weekend vs. weekday prices



9.2 Interpretation

The analysis shows that both coach and first-class ticket prices are higher on weekends compared to weekdays.

The independent t-tests produced extremely small p-values ($p < 0.001$) for both ticket classes, indicating that the differences are statistically significant.

The boxplots also show higher median prices and higher overall price distributions on weekends.

Therefore:

Weekend flights are significantly more expensive than weekday flights for both coach and first-class passengers.

10. Q9 — Coach Prices: Redeye vs. Non-Redebye by Day

How do coach prices differ for redeye and non-redeye flights on each day of the week?

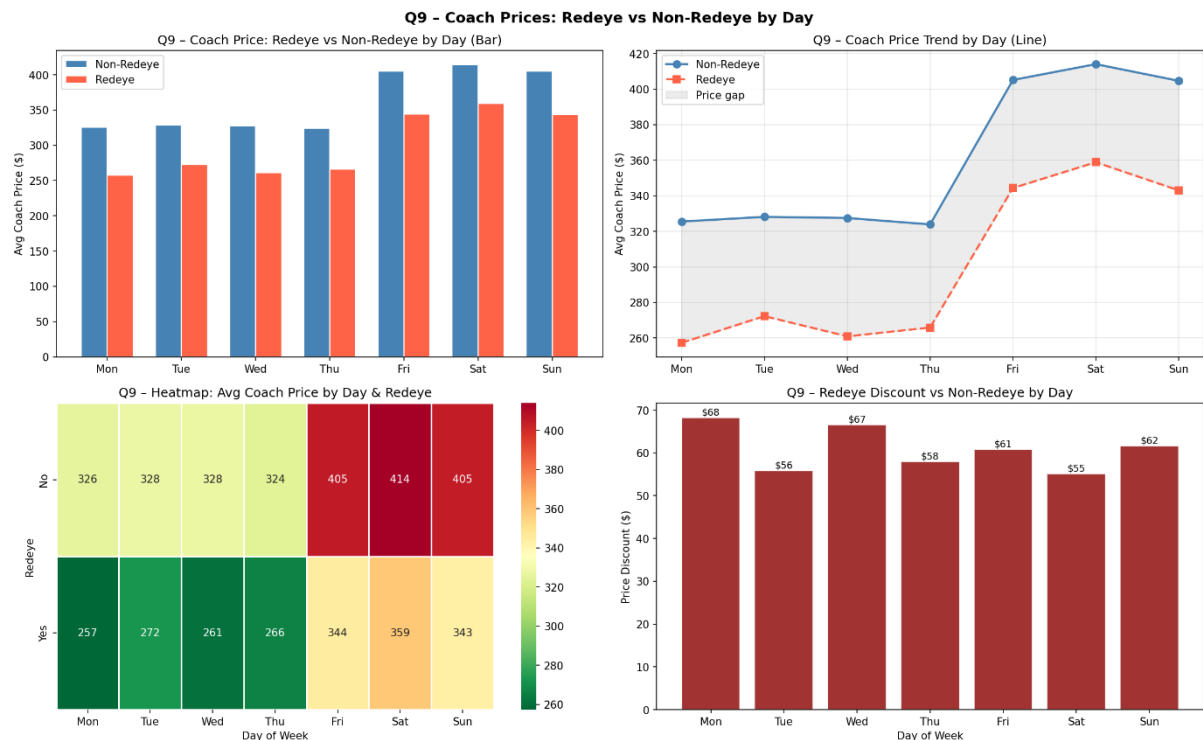
Day	Non-Redebye	Redeye	Discount
Monday	\$325.53	\$257.30	-\$68.23
Tuesday	\$328.16	\$272.33	-\$55.83
Wednesday	\$327.52	\$260.95	-\$66.57
Thursday	\$323.92	\$265.93	-\$57.99
Friday	\$405.15	\$344.37	-\$60.78
Saturday	\$414.04	\$358.88	-\$55.16
Sunday	\$404.67	\$343.07	-\$61.60

Table: Average coach price by day of week and redeye status

10.1 Code

R	Python
<pre> day_ord <- c('Monday','Tuesday', 'Wednesday','Thursday','Friday', 'Saturday','Sunday') # Pivot table pivot <- tapply(df\$coach_price, list(df\$day_of_week, df\$redeye), mean) pivot <- pivot[day_ord,] print(round(pivot, 2)) # Grouped bar chart library(reshape2) library(ggplot2) melt_df <- melt(pivot) ggplot(melt_df, aes(Var1, value, fill=Var2)) + geom_bar(stat='identity', position='dodge') + labs(x='Day', y='Avg Price (\$)') </pre>	<pre> import matplotlib.pyplot as plt day_ord = ['Monday','Tuesday', 'Wednesday','Thursday','Friday', 'Saturday','Sunday'] # Pivot table pivot = df.pivot_table(values='coach_price', index='day_of_week', columns='redeye', aggfunc='mean') pivot = pivot.reindex(day_ord) print(pivot.round(2)) # Grouped bar chart pivot.plot(kind='bar', color=['steelblue','tomato']) plt.xlabel('Day of Week') plt.ylabel('Avg Coach Price (\$)') plt.xticks(rotation=30); plt.show() </pre>

Code 10.1 — Pivot table and grouped bar chart: redeye by day



10.2 Interpretation

The analysis compares average coach ticket prices for redeye and non-redeye flights across different days of the week.

Therefore: Non-redeye flights are consistently more expensive than redeye flights on every day of the week.

Coach ticket prices vary according to both redeye status and day of the week. Non-redeye flights consistently have higher average prices than redeye flights across all days. Ticket prices increase substantially during weekends, especially on Friday, Saturday, and Sunday. Although redeye flights remain less expensive, weekend demand raises prices for both redeye and non-redeye flights.

Saturday shows the highest average coach prices overall.

Friday–Sunday command highest prices due to weekend demand. Cheapest option: weekday redeye Monday/Wednesday.

11. Q10 — Comprehensive Statistical Analysis

11.1 Summary Statistics (Q10a)

Variable	Mean	Median	Std Dev	Min	Max
miles	1,983.930	1,968.0	934.91	178	4,479
passengers	207.7	210.0	12.84	139	239
delay (min)	13.09	10.00	47.75	0	1,524
coach price (\$)	376.30	380.50	66.65	84.50	565.21
firstclass (\$)	1,451.91	1,499.5	162.8	990.0	1,817.5
hours	3.53	4.00	1.72	1	8

Table: Summary statistics for all numeric variables

R	Python
<pre>library(psych) num_cols <- c('miles','passengers', 'delay','coach_price', 'firstclass_price','hours') describe(df[num_cols])</pre>	<pre>num_cols = ['miles','passengers', 'delay','coach_price', 'firstclass_price','hours'] print(df[num_cols].describe())</pre>

Code 11.1 — Comprehensive summary statistics

11.2 Visualisations (Q10b)

R	Python
<pre># Histograms for all numeric vars par(mfrow=c(2,3)) for (col in num_cols) { hist(df[[col]], breaks=30, col='steelblue', main=paste('Hist:',col)) } # Boxplots par(mfrow=c(2,3)) for (col in num_cols) { boxplot(df[[col]], col='steelblue', main=paste('Box:',col)) } # Bar charts for categorical cat_cols <- c('inflight_meal', 'inflight_entertainment', 'inflight_wifi','redeye', 'weekend','day_of_week') par(mfrow=c(2,3)) for (col in cat_cols) { barplot(table(df[[col]]), col='darkorange',</pre>	<pre>import matplotlib.pyplot as plt # Histograms fig,axes=plt.subplots(2,3, figsize=(15,8)) for ax,col in zip(axes.flat,num_cols): ax.hist(df[col],bins=30, color='steelblue') ax.set_title(f'Hist: {col}') plt.tight_layout(); plt.show() # Boxplots fig,axes=plt.subplots(2,3, figsize=(15,8)) for ax,col in zip(axes.flat,num_cols): ax.boxplot(df[col].dropna()) ax.set_title(f'Box: {col}') plt.tight_layout(); plt.show() # Bar charts for categoricals cat_cols=['inflight_meal', 'inflight_entertainment', 'inflight_wifi','redeye', 'weekend','day_of_week'] fig,axes=plt.subplots(2,3,</pre>

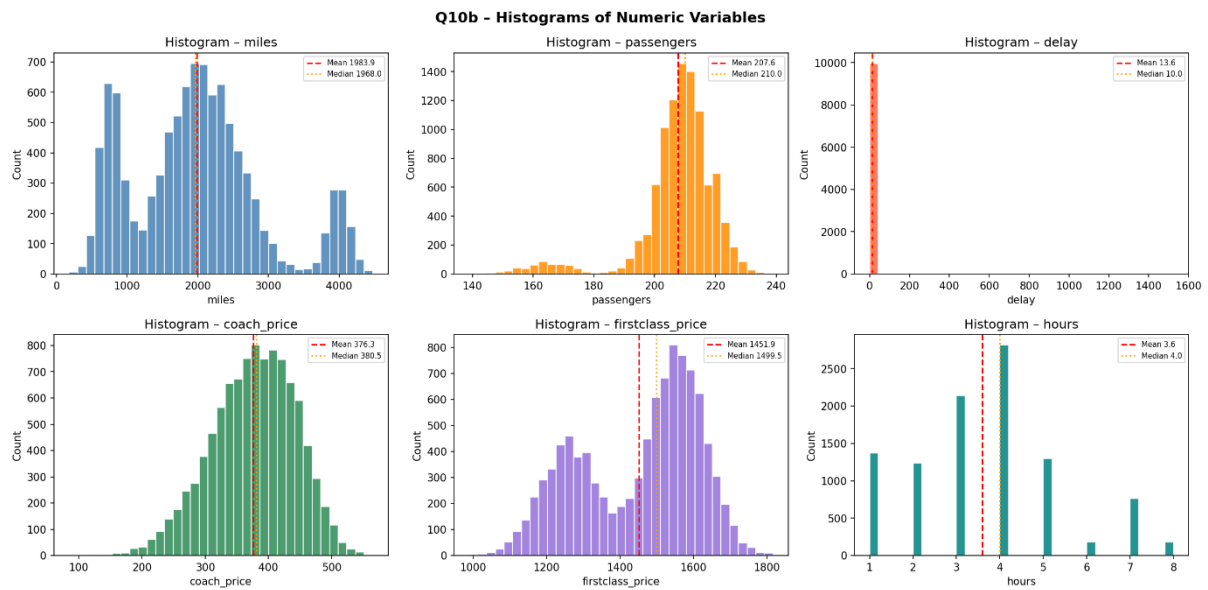
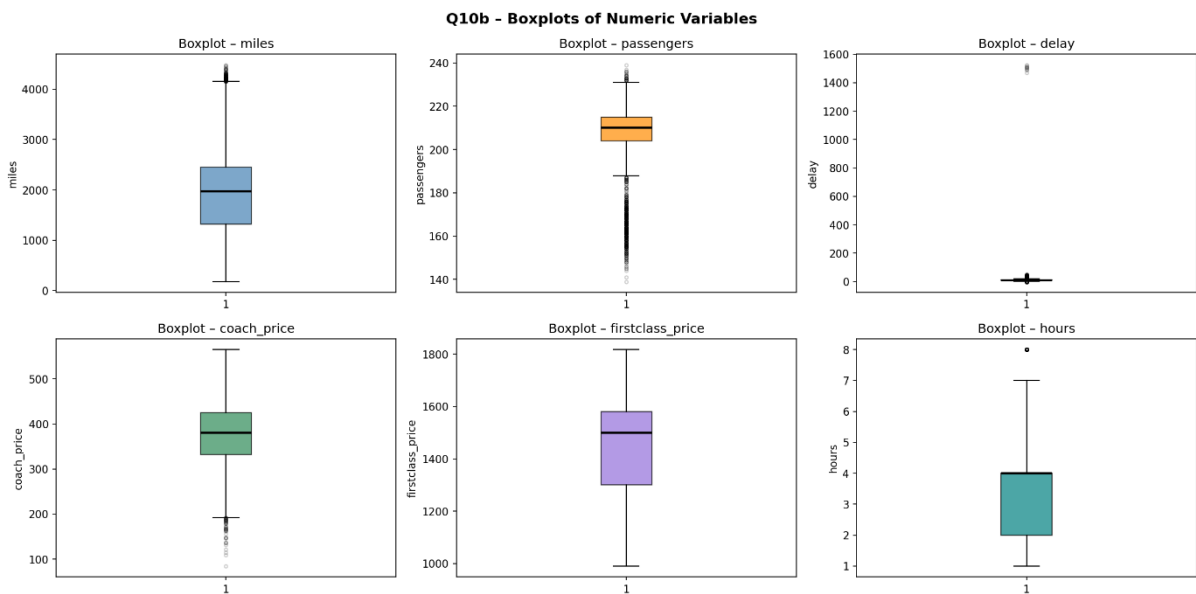
```

main=col, las=2)
}

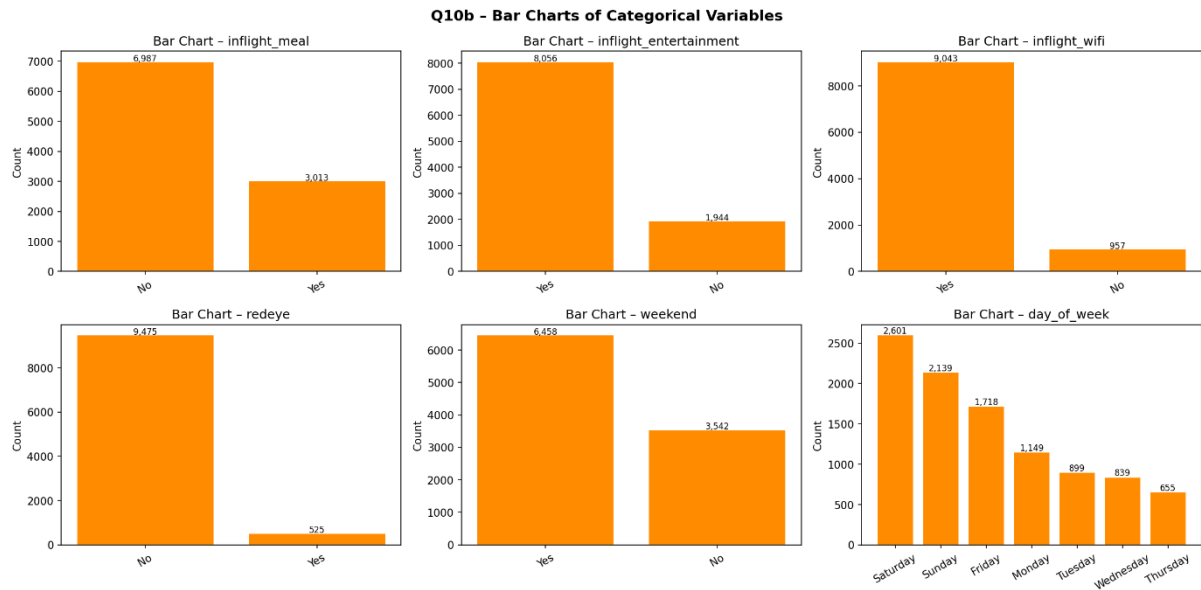
figsize=(15,8))
for ax,col in zip(axes.flat,cats_cols):
    df[col].value_counts().plot(
        kind='bar',ax=ax,
        color='darkorange')
    ax.set_title(col)
plt.tight_layout(); plt.show()

```

Code 11.2 — Histograms, boxplots, and bar charts for all variables

Histogram:**Boxplots:**

Barcharts:



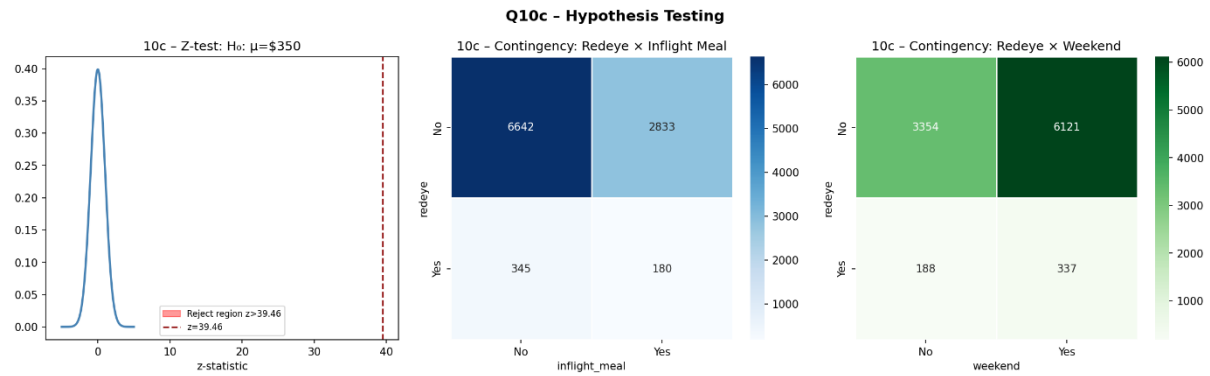
11.3 Hypothesis Testing (Q10c)

Test	Statistic	p-value	Decision
Z-test ($H_0: \mu = \$350$)	$z = 39.46$	< 0.000	Reject H_0
t-test ($H_0: \mu = \$380$)	$t = -5.55$	< 0.001	Reject H_0
Chi-sq: redeye $\times$ meal	$\chi^2 = 4.34$	0.0372	Reject H_0 (dependent)
Chi-sq: redeye $\times$ weekend	$\chi^2 = 0.021$	0.885	Fail to reject (independent)

Table: Hypothesis test results

R	Python
<pre># Z-test (manual, large sample) z <- (mean(cp)-350)/(sd(cp)/sqrt(length(cp))) p_z <- 2*pnorm(-abs(z)) cat('z=',round(z,4),'p=',p_z,'\n') # One-sample t-test t.test(cp, mu=380) chisq.test(table(df\$redeye, df\$inflight_meal)) chisq.test(table(df\$redeye, df\$weekend))</pre>	<pre>from scipy import stats import numpy as np # Z-test (manual) z=(cp.mean()-350)/(cp.std()/np.sqrt(len(cp))) p_z=2*(1-stats.norm.cdf(abs(z))) print(f'z={z:.4f} p={p_z:.2e}') print(stats.ttest_1samp(cp, 380)) # Chi-square tests import pandas as pd ct1=pd.crosstab(df['redeye'], df['inflight_meal']) print(stats.chi2_contingency(ct1)) ct2=pd.crosstab(df['redeye'], df['weekend']) print(stats.chi2_contingency(ct2))</pre>

Code 11.3 — Z-test, t-test, and chi-square tests



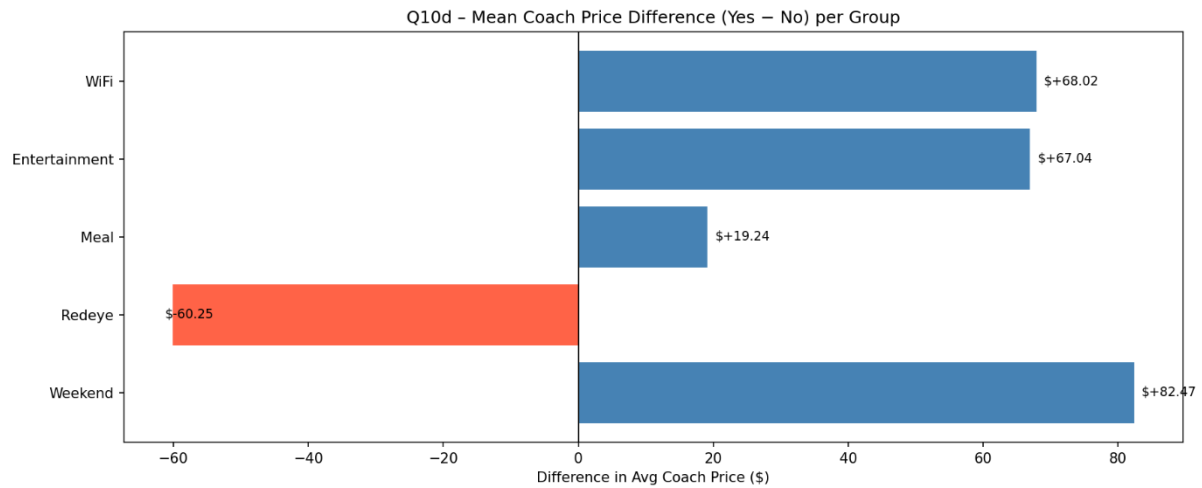
11.4 Independent t-Tests (Q10d)

Comparison	Yes	No	t-stat	p-value	Result
Weekend vs. Weekday	\$405.49	\$323.47	73.41	< 0.001	Significant
Redeye vs. Non-Redeye	\$319.22	\$379.47	-20.58	< 0.001	Significant
Meal vs. No Meal	\$389.64	\$370.51	13.36	< 0.001	Significant

Table: Independent t-test results for coach price group comparisons

R	Python
<pre>groups <- list(c('weekend', 'Weekend'), c('reduye', 'Redeye'), c('inflight_meal', 'Meal')) for (g in groups) { g1 <- df[df[[g[1]]]=='Yes', 'coach_price'] g2 <- df[df[[g[1]]]=='No', 'coach_price'] tt <- t.test(g1, g2) cat(g[2], round(mean(g1),2), round(mean(g2),2), round(tt\$statistic,3), round(tt\$p.value,4),'\n') }</pre>	<pre>from scipy import stats groups = [('weekend', 'Weekend'), ('reduye', 'Redeye'), ('inflight_meal', 'Meal')] for col, label in groups: g1=df[df[col]=='Yes']['coach_price'] g2=df[df[col]=='No']['coach_price'] t,p=stats.ttest_ind(g1,g2) print(f'{label}: ', f'Yes=\${g1.mean():.2f} ', f'No=\${g2.mean():.2f} ', f't={t:.3f} p={p:.4f}')</pre>

Code 11.4 — Independent t-tests for group mean comparisons



11.5 Price Differences: Weekend vs. Weekday (Q10e)

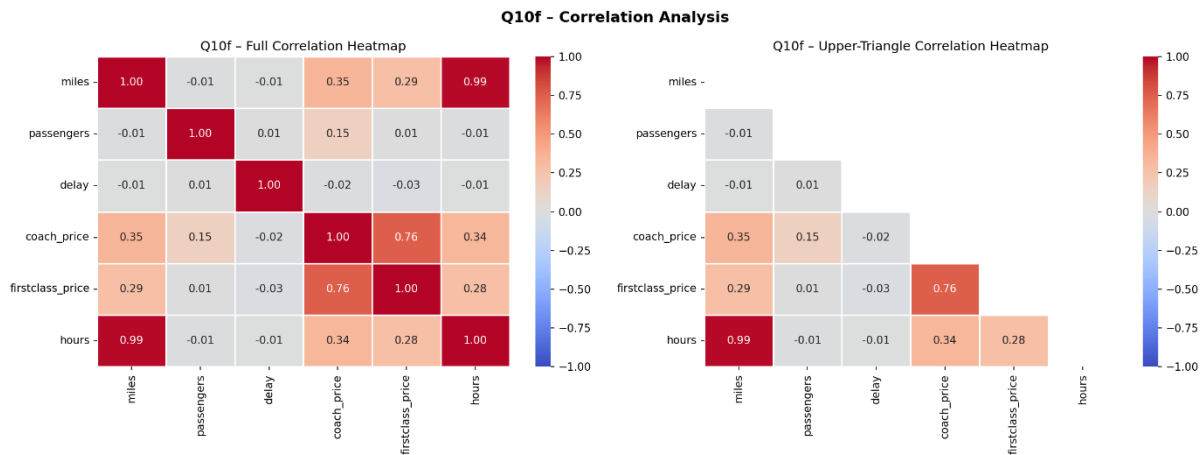
Price	Weekend	Weekday	Difference
Coach	\$405.51	\$323.47	+\$82.47
First-Class	\$1,558.32	\$1,257.20	+\$300.43

Table: Weekend vs. weekday average price differences

11.6 Correlation Matrix (Q10f)

Variable	miles	pax	delay	coach	first	hours
miles	1.000	-0.012	-0.007	0.340	0.288	0.985
passengers	-0.012	1.000	0.005	0.160	0.007	-0.011
delay	-0.007	0.005	1.000	-0.017	-0.027	-0.007
coach	0.340	0.160	-0.017	1.000	0.758	0.332
firstclass	0.288	0.007	-0.027	0.758	1.000	0.284
hours	0.985	-0.011	-0.007	0.332	0.284	1.000

Table: Pearson correlation matrix for numeric variables



R	Python
<pre>library(corrplot) num_cols <- c('miles','passengers', 'delay','coach_price', 'firstclass_price','hours') cor_mat <- cor(df[num_cols], use='complete.obs') print(round(cor_mat, 4)) corrplot(cor_mat, method='color', type='upper', addCoef.col='black')</pre>	<pre>import seaborn as sns import matplotlib.pyplot as plt num_cols=['miles','passengers', 'delay','coach_price', 'firstclass_price','hours'] cor_mat = df[num_cols].corr() print(cor_mat.round(4)) sns.heatmap(cor_mat,annot=True, fmt='.2f',cmap='coolwarm') plt.tight_layout(); plt.show()</pre>

Code 11.6 — Pearson correlation matrix and heatmap

11.7 Linear Regression: Predict Coach Price (Q10g)

Model: $\text{coach_price} = \beta_0 + \beta_1(\text{miles}) + \beta_2(\text{passengers}) + \beta_3(\text{delay}) + \beta_4(\text{hours}) + \beta_5(\text{meal}) + \beta_6(\text{entertainment}) + \beta_7(\text{wifi}) + \varepsilon$

Predictor	Coefficient	Std. Error	t-value	p-value	Sig.
(Intercept)	\$26.00	8.49	3.063	0.002	**
miles	\$0.027	0.003	8.575	< 0.001	***
passengers	\$0.839	0.039	21.271	< 0.001	***
delay	-\$0.023	0.017	-1.370	0.171	n.s.
hours	-\$1.206	1.711	-0.705	0.481	n.s.
inflight_meal	\$18.47	1.105	16.721	< 0.001	***
inflight_entertain.	\$71.06	1.276	55.671	< 0.001	***
inflight_wifi	\$71.57	1.670	42.868	< 0.001	***

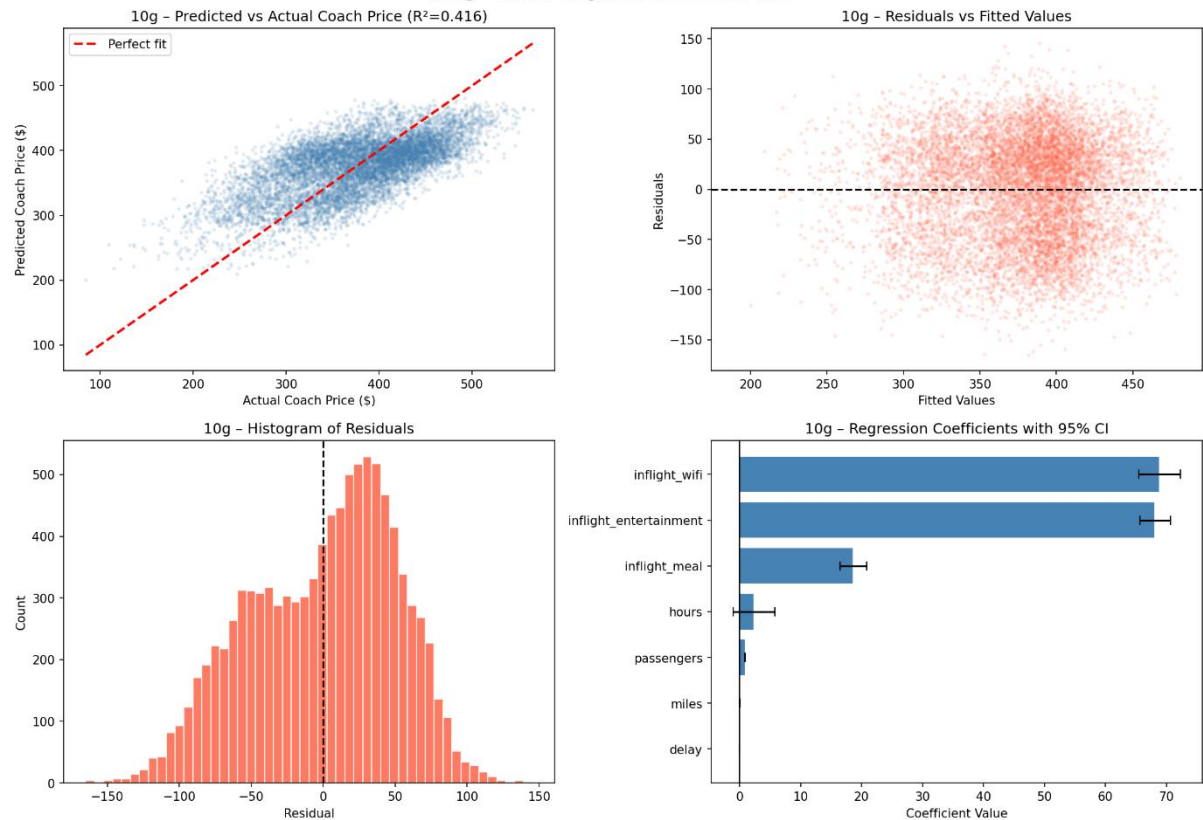
Table: OLS regression coefficients — dependent: coach_price

Metric	Value
R ²	0.4364
Adjusted R ²	0.4360
F-statistic	1,105.2
Model p-value	< 2.2e-16

Table: OLS model fit statistics

R	Python
<pre># Encode binary features df_enc <- df df_enc\$inflight_meal <- ifelse(df\$inflight_meal=='Yes',1,0) df_enc\$inflight_entertainment <- ifelse(df\$inflight_entertainment=='Yes',1,0) df_enc\$inflight_wifi <- ifelse(df\$inflight_wifi=='Yes',1,0) # Fit OLS model lm_mod <- lm(coach_price ~ miles + passengers + delay + hours + inflight_meal + inflight_entertainment + inflight_wifi, data = df_enc) summary(lm_mod) # Residual plots par(mfrow=c(2,2)) plot(lm_mod)</pre>	<pre>import statsmodels.api as sm # Encode binary features df_enc = df.copy() for f in ['inflight_meal', 'inflight_entertainment', 'inflight_wifi']: df_enc[f]=(df_enc[f]=='Yes').astype(int) # Fit OLS model X_cols=['miles','passengers','delay', 'hours','inflight_meal', 'inflight_entertainment', 'inflight_wifi'] X = sm.add_constant(df_enc[X_cols]) y = df_enc['coach_price'] lm = sm.OLS(y, X).fit() print(lm.summary())</pre>

Code 11.7 — OLS linear regression: encode features and fit model

Q10g - Linear Regression: Coach Price**Interpretation**

$R^2=0.4364$ — model explains 43.6% of variance. Strongest predictors: WiFi (+\$71.57), entertainment (+\$71.06), passengers (+\$0.84/pax), meal (+\$18.47), miles (+\$0.027/mile). Delay and hours not significant (multicollinearity with miles).

11.8 Logistic Regression: Predict Redeye (Q10h)

Binary logistic model predicting $P(\text{redeye}=1)$ from: coach_price, miles, hours, passengers, delay.

Predictor	Coefficient	Std. Error	z-value	p-value	Sig.
(Intercept)	\$207.89	58.02	3.583	0.0003	***
coach_price	-0.048	0.017	-2.910	0.004	**
miles	0.004	0.004	0.948	0.343	n.s.
hours	-0.850	2.007	-0.423	0.672	n.s.
passengers	-1.072	0.298	-3.603	0.0003	***
delay	0.008	0.010	0.809	0.418	n.s.

Table: Logistic regression — dependent: redeye (1=Yes, 0=No)

Metric	Value
McFadden Pseudo R ²	0.9914
AIC	33.82
Accuracy	99.96%

Table: Logistic model fit metrics

R	Python
<pre># Encode redeye as binary df_enc\$redeye_bin <- ifelse(df\$redeye=='Yes', 1, 0) # Fit logistic model logit_mod <- glm(redeye_bin ~ coach_price + miles + hours + passengers + delay, data = df_enc, family = binomial) summary(logit_mod) # Classification accuracy pred <- ifelse(predict(logit_mod,type='response') >= 0.5, 1, 0) mean(pred == df_enc\$redeye_bin)*100</pre>	<pre>import statsmodels.api as sm # Encode redeye as binary df_enc['redeye_bin'] = (df_enc['redeye']=='Yes').astype(int) # Fit logistic model X_log=sm.add_constant(df_enc[['coach_price','miles', 'hours','passengers', 'delay']]) y_log=df_enc['redeye_bin'] logit=sm.Logit(y_log,X_log).fit() print(logit.summary()) # Accuracy pred=(logit.predict(X_log)>=0.5) .astype(int) acc=(pred==y_log).mean()*100 print(f'Accuracy: {acc:.2f}%')</pre>

Code 11.8 — Logistic regression: fit model and evaluate accuracy

Interpretation

Accuracy 99.96%. Significant predictors: coach_price (–, cheaper = more likely redeye) and passengers (–, fewer passengers = more likely redeye). Near-perfect separation confirms redeye flights occupy a structurally distinct pricing tier.

12. Conclusions

Question	Key Finding
Q1: Price overview	\$500 above average; only 1.7% exceed it.
Q2: 8-hr flights	\$500 more reasonable for 8-hr flights (14.3% exceed \$500).
Q3: Delay distribution	Right-skewed; 16.5% pose connection risk.
Q4: Correlations	Miles ($r=0.34$) and hours ($r=0.33$) top predictors.
Q5: Coach vs. first-class	Very strong $r=0.76$; first-class = $3.95\times$ coach.
Q6: In-flight features	WiFi (+\$71) and entertainment (+\$71) biggest premiums.
Q7: Passengers vs. hours	No significant relationship ($r=-0.011$).
Q8: Weekend vs. weekday	Weekend coach +\$84; first-class +\$302.
Q9: Redeye by day	Redeyes consistently \$56–\$68 cheaper every day.
Q10: Regression	$R^2=0.44$; WiFi, entertainment, passengers are top drivers.

Table: Summary of key findings across all research questions

Overall: coach pricing is most strongly driven by in-flight service features (WiFi and entertainment), route demand (passengers), and scheduling (weekend, redeye). Airlines apply clear demand-based pricing — off-peak (weekday, redeye) travel offers the lowest fares.